To you who are reading, hello!
My name is Beatrice Occhiena and I am a student of Artificial Intelligence.

Since I study so many different subjects on my own, I tend to rely a lot on all the resources I find on the web 🌐. Whenever I do some research online to learn something new, I always feel very lucky, because I find so much free material carefully crafted by other people from all over the world. So I too decided to make my own small contribution ✨.

Over the months I have put a lot of care into the creation of these notes ✍🏻. And although I know that they may be incomplete and imperfect in some places, I would like to make them available to you — in the hope that they may help even one student in the exam preparation.
I firmly believe in collaboration and mutual support between students 🤝🏻. Furthermore, in the future I would love to be involved in research and education, so this is a first opportunity for me to showcase my work.

That said, I wish you all the best for this exam! Take these words as a small gesture of encouragement. Even though we don't know each other, I still want to cheer you on! Do your best and don't forget to have as much fun as possible studying this subject 💫.

Contact me!

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What you will find in this pdf:

- *IPv4*
- *IPv6*
- *Transition v4 to v6*
- *Wireless networks*
- *Cellular networks*
- *Routing techniques*

FROM DS HOSTS → v4

NETWORK CENTERED SOLUTIONS

Address Family Transit Router (AFTR)
TUNNEL CONCENTRATOR + CAN FEATURE A VERY LARGE NAT

DS-Light "Double Stack"
Not Scalable Solution Since the NAT is not on the CPE
⇩
2 ↑ private net can have = @

A+P "Address + Port"
Every NAT needs a ≠ IPv4@!
⇩
Same IPv4@ & by customers, assigning ≠ TCP/UDP port range

MAP "Mapping Address Port"
No more ranges but SETS of port
⇩
STATELESS config: ↓ customer_ID(ports) w/ the same PSID deduced from network info
⇩
MAPPING : IPv4 in IPv6 unique & topologic...

BEST-SOLUTION

FROM v6 HOSTS → v4

NAT64 + DNS64

④ Host asks for "SITE" IPv6@ to its DNS. The DNS64 asks to the DNS4 the IPv4@.

② The DNS64 : • update the NAT64 w/ info IPv6@(site) ↓ IPv4@(site)
• reply w/ IPv6@ to the HOST

③ The packet is sent & processed by NAT to reach the dest.

NETWORK CENTERED SOLUTIONS

6to4
For NETv6 w/ no special mapping, NO READY ANYWAY

4 v6 v6 — 6 v4 v4

TRANSITION

Why?
#1) Addressing Space
#2) Inefficiency PRIVATE IPv4
#3) Security
#4) Multicast + Anycast
#5) QoS

IPv4 → IPv6

Slowly BECAUSE

- incremental
- smooth

technologies involved

@ MAPPING
IPv4@ + IPv6@ (MAP, Rules)

TRANSLATION
NAT v4/v6 Header Exchange

DUAL-STACK
tecnology for Tot TRANSLATION
IPv6 as a new L3 protocol

TUNNELING (ENCAPSULATION)
protocols:
- GRE
- IPv6 in IPv4(6t4)

Set up:
{6over4, 6to4, Tunnel Broker (TSP, TAIAP, Teredo)}

LAYER 2 DEVICES

SWITCH — DATA GRAM
WI-FI ACCESS POINT — DATA GRAM

direct connection
NO @ IP

LAYER 3 DEVICE

PORT1 IP@1
PORT2 IP@2

A port of a DEVICE capable to connect w/ NETWORK

wi-fi interface IP@1
cellular // IP@2
blue tooth // IP@3

Hosts & Routers INTERFACES

identify

IP address
32-bit IDENTIFIER
NETWORK part | HOST part
PHYSICAL-LINKED HOSTS

original classification

CLASSES @ING

DEN network
A: 0 | B1 B2 B3 B4
B: 10
C: 110
MULTICAST D: 1110 — 224 ↔ 239, reserved
E: 1111

VERY RIGID SCHEME!
Rapid depletion of IP@

Need more than 268 hosts? You get > 60k !!! UNUSED

IANA
Hierarchical ADDRESSING
net ID x.y.0.0 — net ID x.y.z.0
NOT ACCEPTABLE for ROUTING strategies
net ID a.b.c.0/24 — net ID a.b.c.d/28

to overcome ADDRESS EXHAUSTION
OR:

"Class InterDomain Routing"
CIDR method
SUBNET MASKING with VARIABLE LENGTH
/xx → SUBNET MASK of XX-bit of 1's
/24 → 225.225.225.0 NetMask notation
VALID 130.112.1.16/28 — 0004 0000
INVALID 130.112.1.4/28 — 0000 0100

AGGREGATION
to summarize into one single comprehensive network
AGGREGATED in: subnet, supernet

SPECIAL IP@

xxx 000...0	Subnet ID
111 1	local net broadcast
xxx 111...1	broadcast to net xxx
127	loopback

COMMUNICATION

LOCAL NETWORK (LAN) — X.X.X.4, X.X.X.1, X.X.X.2, X.X.X.3
Configuration: IP@ xxx.3, netmask xxx0, default gateway xxx4, DNS yyyy

DIRECT DELIVERY
dest@ = ////// AND host netmask = xxx0 (IEC = host netw.)
ARP protocol to find MAC@ of the ■

INDIRECT
dest@ = ////// AND host netmask ≠ xxx0 (IEC ≠ host mach.)
ARP protocol to find MAC@ of DEFAULT GATEWAY

lv. 2 Comm.

IEEE 802
Group of standards for LAN & MAN
> ex Ethernet, Wifi, ...

ARP LAYER 2/3
"Address Resolution Protocol"

A wants to contact IP@ xxxx.
• ∈ LAN : find its MAC@
• ∉ LAN : find DG's MAC@

ARP CACHE — TO STORE RECEIV. RESULTS

DATA LINK PACKET

MACs	MACA	Payload
REQ.	MACA	BROAD Who is ...3? broadcast
REPLY	MACC	MACA I am ...3! unicast

other technique

MULTICAST
what's up ??
e.g. VIDEO STREAMING, GAMING
GROUP identified by a CLASSD (1110...) IP@
dest@ = GROUP@
HOSTS DYNAMICALLY JOIN & LEAVE

IEEE 802 NETWORK
CONVERSION
1110 MAC@ 01-00-5E-00
28 bits
Spread GROUP

BEYOND LAN
Additional protocols needed

IGMP LAYER 2/3
"Internet Group Management P°"
PC_1's local must possess its gpm
send to gpm IP@ 224.....3
no host

TREE DISTRIBUTION
ISP
No Loops
NOT TRAFFIC FRIENDLY!

ROUTING

LOOK UP TABLE	
DEST	OUTPUT
xxx0/16	2
xx00/16	2
yyyy0/23	3
...	...
dest@	1

NOT ONLY NEXT HOP!

Wow, the router apply the NETMASK of each dest to the DATAGRAM dest@.

Searching for the LONGEST MATCH!
/20 match
/23 match ← WIN!

/20 on dest@ 32 bits
= MATCH

ADDRESS PLAN
Given a certain range of @s
e.g. 40.0.0.0/16

STANDARD METHOD:
• #IP@ ∀ net ⇒ /dim
• ASSIGN starting from LARGER

350 hosts
1 router port
4 special @

120 hosts
1 router port
2 special @

2 router ports
2 special @

R.T. CONFIGURATION
D= direct routing (NEXT_HOP_DEST, PORT_IP@)
S= indirect destination (DEST, NEXT_HOP_DEST)
0.0.0.0/0 DEFAULT @ : if the dest doesn't match anything else, go in that dir.

OB: Select the NEXT HOP for the ONES w/ the 1ST MINIMUM COST

Modifying @ use data
IPD/□ encrypting data & hash (NAT & NAT)
S: 0.0.0.0/0 — AU.IP@

Very Host Automatized in the web context

IP address
128-bit IDENTIFIER
hexadecimal
NETWORK PREFIX | HOST INTERFACE
new protocols

types of IPv6@

UNICAST
6 types of which more important:

extra

- GLOBAL UNICAST — 2 or 3 — ≃ IPv4 public @ — UNIQUE & TOPOLOGY BASED — GLOBAL ROUTING TABLE — ISP — IANA
- UNIQUE LOCAL — FC00 — They can be routed, but not in the global internet — ≃ IPv4 private @ — 40b randomly generated — very low % of duplicates
- LINK LOCAL — FE80 — Designed for packets that will not leave the LINK (≤ LAN) — discarded by routers — For auto@ configuration & neighbor discovery — { Autogenerated from MAC@ }
- SITE LOCAL — FEC0 — Deprecated but still used
- LOOPBACK — 0.....01 — Don't leave the interface ⇒ test purpose
- IPV4 EMBEDDED — 00.....0 FFFF — To aid the transition 4-6 — IPv4@ — v4 v6 v4
- UNSPECIFIED — :: = all 0s — Used in DAD procedure

MULTICAST
Structure: FF type SCOPE GROUPID (4 bits | 4 bits)
How BROADLY to propagate the packet
- WELL-KNOWN — reserved to reach specific hosts
- TRANSIENT — dynamically assigned by multicast apps
- SOLICITED NODE — ≈ to broadcast for neighbor discovery ← denied by EVERY UNICAST @

ANYCAST
Each @ identify a group of devices (≠ MULTICAST)
↳ but only send to the NEAREST! (e.g. DNS)
Difficult to implement

HEADER

TTL — Protocol: indicates the type of the next header encaps. • ICMPv6 • TCP • UDP • SECURITY INFO • DUMMYDATA • @ HOST

PACKET FORMAT

IPv4 — IPv6 — Next Header — EXTENSION HEADERS — Data — to CARRY extra info

- FRAGMENTATION — NOT ALLOWED
- CHECKSUM — EXP. OPT. EXTENDED to PROTOCOLS (TCP/UDP)
- HEADER LENGTH — Fixed 40o

In IPv4 all the options are in the header & must be processed every time
NOW ONLY when NEEDED

DEVICE CONFIGURATION

- @ PREFIX
- @ INTERFACE ID
- DEFAULT GATEWAY
- DNS SERVER

TOTAL MANUAL	STATELESS AUTOCONFIG "SLAAC"	STATEFUL CONFIG
	ICMPv6 → Autogen. → net PREFIX + INT_ID	DHCP "Dynamic Host Config Protocol" → req → IP ok? → safe

ROUTING

Both w/ LONGEST PREFIX MATCH
↳ Hierarchical aggregation

Protocols

INTEGRATED ROUTING	SHIPS IN THE NIGHT
extend the same pr. {IS-IS, BGP}	duplicate protocol. {STATIC, RIP, IGRP, OSPF}
4,6	v4 v6

Steps ∀ HOST INTERFACE

#1 Router Solicit. → NET prefix ⇒ INT_ID
#2 Router Advert.
#3 Create the complete IPv6@
DAD
#4 Send
#5 Mult. Listener Report — to subscribe to the corresponding IPv6 Mult. Mod@

new features

A MANUAL / STATIC
not for large NETs

B EUI-64 process
"Extended Unique Identifier"
MAC@ is guaranteed to be unique if bit 7 in bit 0
unique identifier ID where the 11th bit is 1
INT_ID: 8 bits | FF FE | 8 bits
the 7th bit is changed to 1

C PRIVACY EXTENSION
round EUI-64 + INT_ID from MAC
NOT function
ongoing
bit 7 changed to 0
very unlikely, but theoretically not unique
STORED FOR NEXT CONFIG
DAD duplicate detection in progress (once)

since ALL the interf of a same Host can be DIF =

SCOPED ADDRESS
XXXXX %12 — 4
XXXXX %13 — 32
XXXXX %19 — 45

CHANGED

Layer
3 • IPv4 >> IPv6
• ICMPv4 ARP + IGMP >> ICMPv6 integrated in

ONLY UPDATED

- DNS: record A (name → IPv4@) record AAAA (" → IPv6@)
- RIP & OSPF
- BGP & IDRP
4 • TCP & UDP
7 • Socket interface

treated as a NEW PROTOCOL
IPv4 or IPv6
⇒ ENABLES DUAL STACK APPROACH

LAYER 2 COMMUNICATION

④ MAC of the INTERFACE pre-built by the producer
② Multicast MAC derived by the EVERY UNICAST and then MULTICAST IPv6@
33-33 last 32b
used in NEIGHBOR DISCOVERY to find out the real MAC & in DAD

IPv4: BROADCAST is NEARBY processed by the NIC (ARP)
IPv6: MULTICAST is FILTERED by the NIC (ICMPv6)
not to be going & beyond the local network
Isolated NODE

ICMPv6

Type | Code | Checksum | Message Body

- Diagnostics → ECHO — dest unreachable, TTL exceeded, Packet too big
- Issue notifications
- Neighbor solicit.
- Neighbor advert.
- Multicast Listenin query
- Mult list. report/leave
- Router solicit.
- Router advertis.

ICMP Neighbor Discovery (former ARP)
new method for @ RESOLUTION
I need the MAC of IPv6(B)
B
Multicast MAC(B)
Here it is!
MAC(B)
Neighbor Solicitation
Neighbor Advertisement

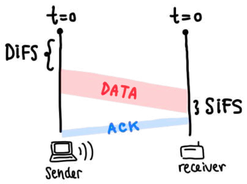
ICMP Multicast Group Management (former IGMP)
Same OB of IGMP
MUL LISTENER QUERY Who is to a (specific) MULTICAST GROUP?
MUL LISTENER REPORT This is my NEW MULT IPv6@!
Wanna LEAVE
Mul. LISTENER DOWN
① to join a group ② NEARBY routers to notify, to inform the routers of its SOCIAL MOVING@

ICMP Router Prefix Discovery
Router Adv. IP id ...
Router Solic. What's your net prefix?
OR
UNSOLICITED: periodically send IP id ...

DAD
"Duplicate Address Detection"
OB: probe the uniqueness of an IPv6@ on a LAN
multicast
I'll take your IPv6@ value!
Src | Src | ICMPv6
UNICAST(x:x:x) ⇒ MULT.COL.MOD (FF...)
① ¼ WAIT for RESPONSE

W/OUT RESERVATION

only **ACK**



Sense FREE before TRANSMITTING

- yes — Go!
- no — Random backoff time

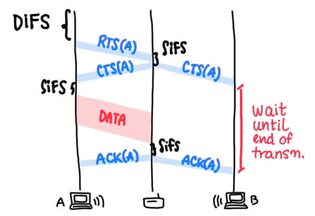
type 1

802.11
CSMA/CA collision avoidance
to avoid signal disruption

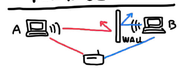
type 2

W/ RESERVATION

RTS-CTS
Request To Send
Clear To Send



HIDDEN TERMINAL PROBLEM



A,B unaware of their INTERFERENCE at the ACCESS POINT because of a barrier

SIGNAL ATTENUATION PROBLEM

A,B too distant to reach each other w/ their signal

but still

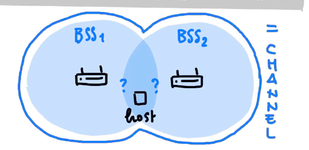
CSMA protocol 802.11

Manage MULTIPLE ACCESS w/ SENSE BEFORE TRANSMITTING tech.

protocols needed to avoid undue privileges for ↓ DIFS

≈ 50 µs
listening for **DIFS** time

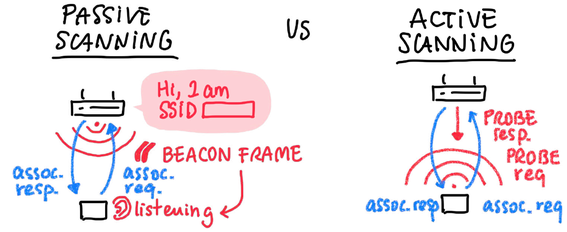
CHANNEL ASSOCIATION



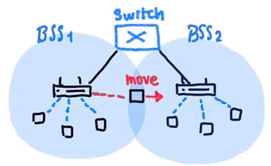
The host must:

1) SCAN the channel
2) Select AP to ASSOCIATE
3) AUTHENTICATION
4) Run DHCP to get an IP@

PASSIVE SCANNING vs ACTIVE SCANNING



MOBILITY WITHIN = SUBNET



the host mantains Same IP@

more difficult comm.

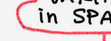
METRICS to measure quality

- **SNR** "Signal-to-Noise Ratio" ↑ Good
- **BER** "Bit Error Rate" ↓ Good

VARIABLE in SPACE & TIME

RATE ADAPTATION

When BER too ↑



switch to ↓ transm.rate

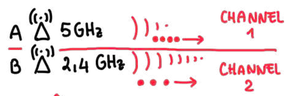
MORE HOSTS can send at the = time

NEIGHBOR AP w/ = channel

FREQUENCY MULTIPLEXING

≠ bands with:

- max transm rate
- max distance



INTERFERENCE

BROADCAST

MULTIPATH PROPAGATION

bouncing of the = signal

PATH LOSS

- strength of signal ↓ $\propto \frac{1}{d^2 \cdot freq^2}$

Both a PROBLEM & DESIRABLE
↓
spatial distribution to ↓ interf.

WIRELESS LINK

regulated by

family of lv2 protocols

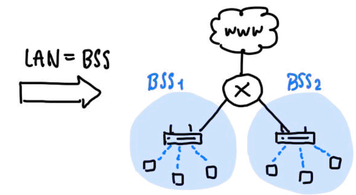
IEEE 802.11 LAYER 2
protocol for WIRELESS LAN

type	MAX bitrate	range	freq
WiFi 4	?	70 m	2,4/5 GHz
WiFi 5	3,7 Gbps	70 m	5 GHz
WiFi 6	14 Gbps	70 m	2,4/5 GHz

FRAME HEADER

type	@1	@2	@3
ACK RTS CTS data	dest	src	AP

LAN = BSS



WIRELESS TECHNOLOGY

extension of

COMPARED to CABLE Connection:

- ERROR ↑
- MOBILITY ↑
- FLUIDITY ↑

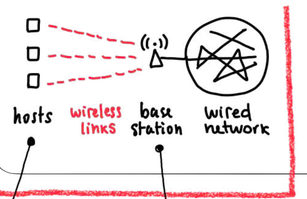
why?

VERY COMMON NOWADAYS

move around

Mode

INFRASTRUCTURE vs AD HOC



hosts only (++ military application)

- mobile (💻📱...)
- stationary (🖥 IoT)

- wifi access point
- cell towers

LAN
AP
node of the subnet
HIGH SPEED
both
on MOBILITY
connected directly to the network
SLOWER SPEED
wide geographical area

CELLULAR NETWORKS

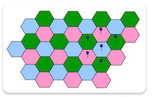
CELLULAR COVERAGE

practical mixed topology based on:
- density of
- morphology
- propagation conditions

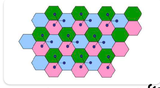
MICRO CELLS
small · many
neighborhood

MACRO CELLS
big · few
rural area

theoretical perfect spatial tassellation w/ hexagonal cells



– isotropic



– 3 directional
(ONLY 1 A ⇒ cost ↓)

to provide

CELL

POWER
HIGH
$$$ TOWER
+
$ ANTENNA
OPERATOR

shared by many

CHANNEL ACCESS

Employ ≠ techniques at the same time:

- FDMA "Freq. Division Multiple Access"
- TDMA "Time —"
- CDMA "Code —" → deployed in 3G, deprecated
- SDMA "Spatial —" → REUSE of FREQ.

REGISTRATION

POWER CONTROL

lower the pwr of the signal!

- ↓ interference
- ↓ pwr consumpt
- ↓ signal intensity
  also good x health

⇒ MORE CELLS = LESS RADIATION

collected in

CLUSTERS

3-cell
Contains All the frequencies E OPERATOR
7-cell
plan
$$$$
GOVERNMENT
limited resources well distributed

↓ dim cell

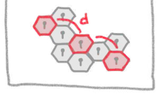
covers N
∀ cell ($\frac{1}{3}$ TOT freq)

in the same area covers M

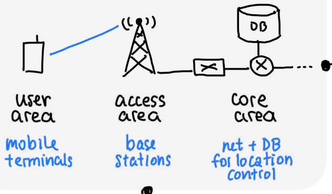
↑ n° cell per cluster

∀ cell, fewer ≠ frequencies
($\frac{1}{7}$ · TOT)

FREQ. REUSE



ARCHITECTURE



MOBILITY MANAGEMENT

LOCATION AREA

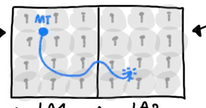
not on cell level (↓ consump.)

ROAMING

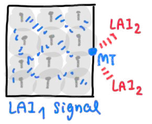
DB
user's position in LA1

PAGING

HANDOVER



LOCATION UPDATING



when MT sense a ≠ LAI

time evolution

Big investments $$$ by European operators

1990	2002	2013	2020
2G GSM (call + sms)	3G UMTS HSPA	4G LTE (call, sms, DATA)	5G

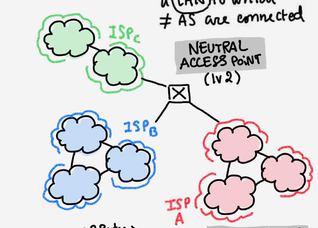
2G – GSM –

4G – LTE –

OB: flexibility to change infrastructure

neutral entity (no $$$ w/ ISP)

a LAN to which ≠ AS are connected

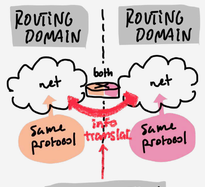


IANA ← 2 Bytes → ID

- AUTON. INNER CHOICHES
- NEGOTIATED EXTERNAL CH.

AUTONOMOUS SYSTEMS

Set of subnets w/ ⊏ routing administr. choiches (e.g. subnets ∈ ISP)



REDISTRIBUTION

translation of coming info from 1 protocol to the other

terminology

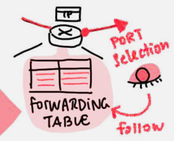
ROUTING

- Ignore TRAFFIC ISSUES ⊘
- Focus on FINDING the "BEST PATH" (shortest / cheaper)
- Determine REACHABLE DEST

"ROUTING" PROACTIVE

Compute BEST ROUTE → Generate → FORWARDING TABLE

"FORWARDING" ON-THE-FLY



algorithms

- by NET (@) — e.g. IP
- LABEL SWAPPING — e.g. MPLS
- SOURCE ROUTING — partial knowledge of the NET

types

protocols

IGP

"Interior Gateway Protocols"

INTRA domain

exchange topological info in the = admin

⇓

OB: find the SHORTEST path

RIP	OSPF
IGRP	IS-IS
DV	LS

FREE STANDARD — 3 min convergence

PROPRIET. SOL. by CISCO — supports MULTIPLE PATH ROUTING

EGP

"Exterior Gateway Protocols"

INTER domain

OB: find the CHEAPEST path

BGP PV

IDRP

Hierarchical Routing

• SUBOPTIMAL

vs

NON ADAPTIVE (static)

STATIC/MANUAL ROUTING

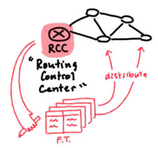
- ⊘ errors ↗
- ⊘ adaptability ↘
- ⊘ back-up route in case of failure?

FLOODING

SELECTIVE FLOODING

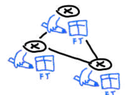
ADAPTIVE (dynamic)

CENTRALIZED ROUTING



- ⊘ single point of failure
- ⊘ bottleneck
- ⊘ not for highly dynamic netw.
- ✓ always best route

ISOLATED ROUTING



- ⊘ no info exchanged!
- ⇒ random forwarding mitigate by backward learning

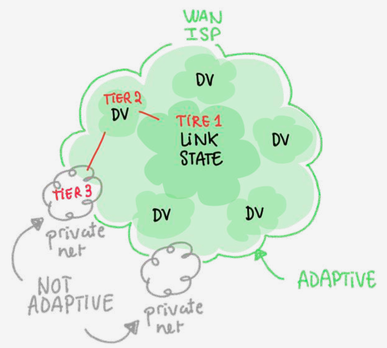
(Not used)

DISTRIBUTED ROUTING

✓ GOOD COMPROMISE

algorithms

NETWORK TIER SUBDIVISION



PATH VECTOR

the same but

Path Vector(N)

dest	dist	path
A	0	[-]
B	1	[A]
C	1	[A]
D	2	[A,B]

⇓

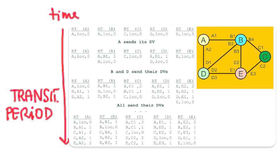
to count to ∞ !!!

DISTANCE VECTOR

algorithm

Bellman-Ford

Simple but slow $O(n^2)$ — overall — for small less dynamic nets

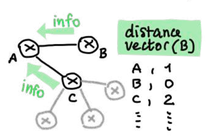


- List of ALL DEST
- Communication just between neighbors
- Propagation in time

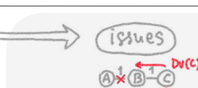
⇒ TRANSITORIES REQUIRED

- Focus on DISTANCE

⇒ IGNORE TOPOLOGY



issues



distance of A continuously recompute

- COUNT TO ∞ — dist(A) → ∞
- BOUNCING EFFECT — ping-pong

Solutions

SPLIT HORIZONS

When a ✕ occours to a neighbor, ⊗ delete its entry from its DV

D&C still think to can reach A from other directions

PATH HOLD DOWN

omitted from list of ALL dest

convergence time ↗

ROUTE POISONING

NOT omitted but dist = ∞

convergence ↑↑↑

LINK STATE

algorithm

Dijkstra

complex but quick — highly dynamic nets!

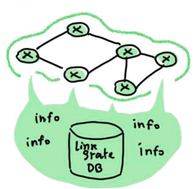
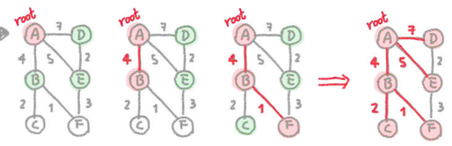
- Info on the STATE of each LINK
- Communication between ALL nodes

⇑ SELECTIVE FLOODING

- local map grows quickly

↳ $\Theta(L \cdot \log(N))$

- SHORTEST PATH FIRST

↳ choose the branch w/ lower cost!

↳ RAISES LOOPS